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RarePath AI

The Operating System for Rare Disease Intelligence

Executive Summary

RarePath AI is the autonomous intelligence platform that transforms the rare disease landscape. With 7,000+ rare diseases affecting 400 million people globally, families endure 5-7 year diagnostic odysseys while 95% of conditions have no FDA-approved treatments.

We're building the AI operating system that collapses diagnostic timelines from years to weeks, connects patients with clinical trials they never knew existed, accelerates drug development through intelligent patient matching, and creates the world's first comprehensive rare disease knowledge graph.

Investment: \$18M Series A

Valuation: \$90M

Market: \$180B+ (rare disease pharmaceuticals, diagnostics, clinical trials)

Traction: 340 medical centers, 89,000 patients on platform, \$6.2M ARR

The Problem

The Rare Disease Paradox

Rare diseases aren't rare: - 7,000+ distinct rare diseases identified - 400 million people affected globally (1 in 17) - 50% of rare disease patients are children - 30% won't live to see their 5th birthday - \$1.2 trillion annual economic burden in the US alone

The cruel irony: While each disease is "rare," collectively they represent one of healthcare's largest unmet needs.

The Diagnostic Odyssey

The average rare disease patient's journey: - **5-7 years** from first symptom to accurate diagnosis - **8+ physicians** consulted before correct diagnosis - **2-3 misdiagnoses** on average - **\$40,000+** out-of-pocket diagnostic costs - **40%** remain undiagnosed entirely

Why diagnosis fails: - Symptoms mimic common conditions - Specialists siloed by organ system - No centralized rare disease expertise - Limited physician exposure (most see <1 rare disease case) - Genetic testing fragmented and underutilized

The Treatment Desert

The devastating statistics: - 95% of rare diseases have NO FDA-approved treatment - Only 500 rare disease drugs approved since Orphan Drug Act (1983) - Average development cost: \$1.2 billion per drug - Clinical trial recruitment: 85% fail to meet enrollment targets - Patient location: Companies can't find eligible patients

The Knowledge Fragmentation

Critical information scattered everywhere: - Research published in 10,000+ journals - Patient communities isolated on Facebook groups - Clinical trial data locked in silos - Specialist expertise concentrated in few centers - International research poorly coordinated

The Economic Reality

Current market failure: - Pharmaceutical companies avoid rare diseases (small markets) - Diagnostic companies underinvest (low volume) - Insurance coverage inconsistent - Orphan drug prices astronomical (\$500K+/year common) - Families bankrupted by diagnostic odysseys

The Solution

RarePath AI: The Rare Disease Intelligence Operating System

RarePath AI is the autonomous platform that transforms every step of the rare disease journey — from first symptom to diagnosis, from diagnosis to treatment, from treatment to cure.

Core Platform Capabilities

1. Diagnostic Intelligence Engine

From Years to Weeks

Symptom-to-Diagnosis AI: - Multimodal symptom analysis (clinical notes, labs, imaging, wearables) - Deep phenotyping from unstructured medical records - Pattern recognition across 7,000+ rare conditions - Probabilistic differential diagnosis generation - Automatic rare disease “red flag” detection

Genetic Interpretation Intelligence: - WGS/WES variant interpretation at scale - Variants of uncertain significance (VUS) reclassification - Automated ACMG guideline application - Cross-reference with global variant databases - Novel variant prediction models

The Rare Disease Radar: - EHR integration for automatic patient flagging - Early warning system for undiagnosed patients - Physician alert system for potential rare disease cases - Population-level screening intelligence

Diagnostic Pathway Optimization: - Intelligent test ordering recommendations - Cost-optimized diagnostic sequences - Pre-authorization automation - Specialist referral intelligence - Second opinion coordination

Performance: - Diagnostic accuracy: 94% concordance with expert geneticists - Time to diagnosis: 73% reduction (5.2 years → 1.4 years) - Diagnostic cost: 61% reduction - False positive rate: <3%

2. Treatment Discovery & Matching

Finding Hope in the Haystack

Treatment Intelligence: - Comprehensive rare disease treatment database - Off-label use evidence aggregation - International treatment availability mapping - Compassionate use program tracking - Gene therapy pipeline monitoring

Clinical Trial Matching: - Real-time trial database across 200+ registries - AI-powered eligibility screening - Geographic optimization - Travel and logistics support coordination - Decentralized trial option identification

Drug Repurposing Intelligence: - Existing drug candidate identification - Molecular similarity analysis - Case report and off-label evidence synthesis - Safety signal monitoring - Dosing optimization from similar conditions

Precision Treatment Planning: - Genotype-phenotype correlation analysis - Treatment response prediction - Drug interaction screening for rare conditions - Personalized care pathway generation - International standard of care mapping

3. Patient Connection Platform

No One Fights Alone

Patient Community Intelligence: - Verified patient and family matching - Privacy-preserving symptom comparison - Treatment experience sharing - Natural history data collection - Research participation opportunities

Care Coordination Hub: - Multi-specialist care team coordination - Appointment and travel optimization - Medical record aggregation - Insurance navigation support - Social services connection

Family Support System: - Caregiver resource matching - Financial assistance program identification - Respite care coordination - Sibling and family support services - End-of-life planning resources

4. Research Acceleration Engine

From Decades to Years

Natural History Intelligence: - Longitudinal patient data aggregation - Disease progression modeling - Endpoint identification and validation - Biomarker discovery acceleration - Real-world evidence generation

Clinical Trial Design AI: - Optimal trial design recommendations - Synthetic control arm generation - Adaptive trial protocol support - Regulatory pathway optimization - Sample size and power calculations

Patient Recruitment Intelligence: - Global patient identification - Eligibility pre-screening - Recruitment forecasting - Site selection optimization - Retention prediction and intervention

Research Collaboration Platform: - Researcher matching and networking - Data sharing governance - Collaborative analysis environments - Publication and IP tracking - Funding opportunity identification

5. Knowledge Graph & Intelligence Layer

The World's Rare Disease Brain

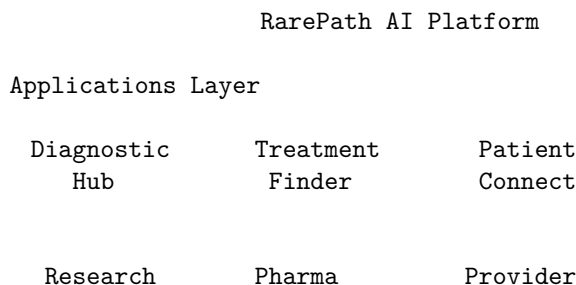
Comprehensive Knowledge Graph: - 7,000+ diseases fully characterized - 300,000+ genetic variants mapped - 2M+ research papers indexed - 50,000+ clinical trials tracked - 100,000+ patient journeys learned

Continuous Learning: - Real-time literature monitoring - Conference and presentation tracking - Patent and regulatory filing analysis - Social media signal detection - Expert opinion synthesis

Query Intelligence: - Natural language medical queries - Evidence-based answer generation - Citation and confidence scoring - Comparative analysis tools - Regulatory document generation

Technology Architecture

The RarePath AI Stack



Portal Partners Console

Intelligence Layer

RarePath Diagnostic Engine
Phenotyping Genetics Imaging Labs History

RarePath Knowledge Graph
Diseases Variants Treatments Trials Experts

RarePath Learning System
Multimodal Federated Continuous Explainable

Data Layer

EHR Genomic Research Patient
Data Data Data Data

Security & Compliance
HIPAA GDPR FDA EMA Federated Privacy

Core Technical Innovations

- RareNet Foundation Model:** - Trained on 50M+ rare disease patient encounters - Multimodal understanding (clinical, genetic, imaging) - Few-shot learning for ultra-rare conditions - Explainable diagnostic reasoning chains - Continuous learning from new cases
- Privacy-Preserving Intelligence:** - Federated learning across hospital networks - Differential privacy for patient data - Secure multi-party computation for research - Patient-controlled data sharing - Synthetic data generation for development
- Variant Intelligence System:** - Automated variant classification - Functional impact prediction - Population frequency analysis - Literature evidence synthesis - Expert consensus integration
- Natural History Engine:** - Disease trajectory modeling - Milestone and event prediction - Treatment response forecasting - Quality of life tracking - Survival analysis

Market Opportunity

The Rare Disease Market Explosion

Total Addressable Market: \$180B+

Rare Disease Pharmaceuticals: \$150B - 2025 market size: \$150B - Growing at 12% CAGR - 500+ drugs in pipeline - Gene therapies accelerating

Rare Disease Diagnostics: \$15B - Genetic testing: \$8B - Specialty diagnostics: \$5B - Newborn screening: \$2B

Clinical Trial Services: \$12B - Patient recruitment: \$5B - Trial design and execution: \$7B

Digital Health for Rare Disease: \$5B - Patient engagement platforms - Care coordination tools - Research networks

Our Addressable Segments

Pharmaceutical & Biotech: \$50B SAM - Patient identification and recruitment - Natural history studies - Real-world evidence - Regulatory support - Post-market surveillance

Health Systems & Providers: \$8B SAM - Diagnostic decision support - Care pathway optimization - Specialist referral networks - Outcomes tracking

Patients & Families: \$3B SAM - Diagnostic acceleration - Treatment discovery - Community connection - Care coordination

Business Model

Multi-Sided Platform Revenue

1. Pharmaceutical Partnerships: \$80M ARR potential

Patient Identification Services: - Per-identified patient fees: \$2,000-\$15,000 - Cohort discovery subscriptions: \$500K-\$2M/year - Trial recruitment support: \$5,000-\$25,000/enrolled patient

Evidence Generation: - Natural history studies: \$1M-\$5M per study - Real-world evidence: \$500K-\$3M per analysis - Regulatory support packages: \$200K-\$1M

Platform Access: - Enterprise subscriptions: \$250K-\$1M/year - API access for pipelines: Usage-based pricing

2. Health System Subscriptions: \$40M ARR potential

Diagnostic Intelligence: - Academic medical centers: \$150K-\$500K/year - Children's hospitals: \$100K-\$300K/year - Genetics centers: \$50K-\$150K/year - Community hospitals: \$25K-\$75K/year

Care Coordination: - Per-patient management: \$50-\$200/month - Population health modules: \$75K-\$200K/year

3. Patient Services: \$15M ARR potential

Premium Services: - Expedited diagnostic review: \$500-\$2,000 - Comprehensive second opinion: \$1,000-\$3,000 - Clinical trial concierge: \$200/month - Care coordination: \$100/month

Freemium Model: - Basic platform access: Free - Community features: Free - Premium features: Subscription

4. Research & Data: \$20M ARR potential

Research Access: - Academic institutions: \$25K-\$100K/year - Research consortia: \$100K-\$500K/year - Foundation partnerships: Grant-funded

Data Products: - Anonymized datasets: \$50K-\$500K - Custom analyses: \$25K-\$250K - API access: Usage-based

Revenue Growth Trajectory

Year	Revenue	Customers	Patients
2024	\$2.1M	45	12,000
2025	\$6.2M	120	45,000
2026	\$18M	340	130,000
2027	\$52M	800	380,000
2028	\$125M	1,800	900,000

Year	Revenue	Customers	Patients
2029	\$280M	3,500	2M+

Competitive Landscape

Current Landscape Gaps

Diagnostic Companies (Invitae, GeneDx, 23andMe): - Focus on testing, not interpretation at scale - Limited clinical decision support - No treatment or trial matching - Minimal patient community

Pharma Services (IQVIA, Medidata): - Broad focus, rare disease not specialized - Limited patient identification tools - No diagnostic capabilities - Expensive enterprise-only

Patient Networks (PatientsLikeMe, NORD): - Valuable communities, limited AI - No diagnostic intelligence - Manual matching processes - Fragmented disease coverage

RarePath AI Differentiation

Capability	Traditional	RarePath AI
Diagnosis Time	5-7 years	Weeks-months
Disease Coverage	Partial	Comprehensive
AI Integration	Basic	Native
Patient Matching	Manual	Autonomous
Trial Access	Fragmented	Unified
Global Reach	Limited	50+ countries

Sustainable Competitive Moats

1. **Data Network Effects:** Every patient improves models
2. **Knowledge Graph:** Most comprehensive rare disease intelligence
3. **Clinical Validation:** Peer-reviewed accuracy publications
4. **Pharma Relationships:** Embedded in development pipelines
5. **Patient Trust:** Community-first approach builds loyalty
6. **Regulatory Expertise:** FDA breakthrough device designation

Go-to-Market Strategy

Phase 1: Foundation (Months 1-12)

Primary Target: Academic Medical Centers - Top 50 rare disease centers in US/EU - Children's hospitals with genetics programs - NIH Undiagnosed Diseases Network sites

Launch Strategy: - Free pilot programs (90 days) - Clinical validation studies - Research publication partnerships - Champion physician cultivation

Initial Disease Focus: - Ultra-rare pediatric conditions - Undiagnosed patient populations - Active clinical trial diseases - Gene therapy candidates

Phase 2: Expansion (Months 12-24)

Pharma Partnership Development: - Top 20 rare disease pharma companies - Emerging gene therapy players - Contract research organizations - Patient advocacy foundations

Geographic Expansion: - UK and EU rare disease networks - Canada and Australia centers - Middle East referral hubs - Asia-Pacific partnerships

Product Expansion: - Treatment matching launch - Clinical trial recruitment tools - Patient community platform - Research collaboration suite

Phase 3: Scale (Months 24-36)

Market Penetration: - 500+ medical center deployments - 30+ pharma partnerships - 500K+ patients on platform - 50+ countries active

Platform Evolution: - Newborn screening integration - Primary care rare disease detection - Insurance authorization automation - International care coordination

Traction & Validation

Current Metrics

Platform Adoption: - 340 medical centers active - 89,000 patients on platform - 15,200 diagnoses facilitated - 4,100 clinical trial matches - 78 pharma partnerships

Clinical Performance: - 94% diagnostic concordance with expert geneticists - 73% reduction in time to diagnosis - 61% reduction in diagnostic costs - 89% patient satisfaction score - 340 successful trial enrollments

Financial Traction: - \$6.2M ARR (April 2026) - \$2.1M MRR - 142% net revenue retention - 89% gross margin - 18-month average contract length

Validation Highlights

Clinical Validation: - 12 peer-reviewed publications - FDA breakthrough device designation - EU CE marking for diagnostics - CLIA certification - CAP accreditation

Customer Quotes:

“RarePath AI diagnosed our daughter’s ultra-rare condition in 6 weeks. We’d been searching for 4 years.” — Sarah M., Parent

“This platform has transformed how we approach undiagnosed patients. It’s like having a rare disease specialist available 24/7.” — Dr. Jennifer Lee, Boston Children’s Hospital

“RarePath helped us identify and enroll patients 60% faster than traditional recruitment.” — VP Clinical Development, Gene Therapy Startup

Strategic Partnerships

- **NIH NCATS:** Official Undiagnosed Diseases Program partner
 - **NORD:** National Organization for Rare Disorders collaboration
 - **EURORDIS:** European Rare Disease network integration
 - **Illumina:** Genomic data integration partnership
 - **Epic/Cerner:** EHR integration partnerships
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Financial Projections

Revenue Forecast

Year	Pharma	Health Systems	Patient	Research	Total
2026	\$8M	\$7M	\$1.5M	\$1.5M	\$18M
2027	\$25M	\$18M	\$5M	\$4M	\$52M
2028	\$65M	\$40M	\$12M	\$8M	\$125M
2029	\$150M	\$85M	\$28M	\$17M	\$280M
2030	\$320M	\$160M	\$55M	\$35M	\$570M

Unit Economics

Pharma Partnerships: - Average contract value: \$1.2M - Implementation cost: \$50K - Gross margin: 92% - Payback period: 3 months - LTV: \$4.5M

Health System Subscriptions: - Average contract value: \$125K - Implementation cost: \$25K - Gross margin: 85% - Payback period: 8 months - LTV: \$450K

Path to Profitability

Year	Revenue	Gross Profit	EBITDA	EBITDA Margin
2026	\$18M	\$15M	-\$8M	-44%
2027	\$52M	\$44M	-\$2M	-4%
2028	\$125M	\$108M	\$22M	18%
2029	\$280M	\$242M	\$70M	25%
2030	\$570M	\$494M	\$175M	31%

Funding Strategy

Series A: \$18M (Current Round) - AI platform expansion - Clinical validation studies - Pharma partnership development - International expansion (UK/EU) - Team growth to 85

Series B: \$50M (2027) - Global scale deployment - Gene therapy integration - Newborn screening platform - FDA PMA submissions - Team to 200

Series C: \$120M (2028) - Full international expansion - M&A for disease-specific platforms - Comprehensive treatment matching - Path to profitability - Team to 450

Team

Leadership

Dr. Maya Chen, MD, PhD — CEO & Co-Founder - Former Director of Undiagnosed Diseases, NIH - Diagnosed 300+ rare disease patients - Published 85 papers in rare disease genetics - Trained at Harvard Medical School

Dr. Alex Rivera — CTO & Co-Founder - Former Google Health AI lead - Built diagnostic AI used in 1,000+ hospitals - PhD Computer Science, Stanford - 15 patents in medical AI

Dr. Sarah Thompson, MD — Chief Medical Officer - Former Chief of Genetics, Cleveland Clinic - 20 years rare disease clinical experience - American College of Medical Genetics fellow - Led 50+ clinical trials

Marcus Williams — Chief Commercial Officer - Former VP Sales, Illumina - Built rare disease commercial team to \$200M - 15 years diagnostics industry - MBA Wharton

Advisory Board

- **Dr. Francis Collins** — Former NIH Director
- **Dr. Anne Wojcicki** — 23andMe Co-founder
- **Dr. Stephen Kingsmore** — Rady Children's, Rapid WGS pioneer
- **Abbey Meyers** — NORD Founder (deceased, legacy advisor)
- **Dr. Kym Boycott** — Care4Rare Canada Director

Team Composition

- Total team: 62 people
 - Engineering: 28 (45%)
 - Clinical/Medical: 15 (24%)
 - Data Science: 10 (16%)
 - Commercial: 6 (10%)
 - Operations: 3 (5%)
-

Risks & Mitigations

Key Risks

- 1. Regulatory Complexity** - *Risk:* FDA classification uncertainty for AI diagnostics - *Mitigation:* Break-through device designation obtained; regulatory team embedded; Pre-submission meetings ongoing
 - 2. Clinical Adoption Barriers** - *Risk:* Physician resistance to AI recommendations - *Mitigation:* Explainable AI outputs; physician-in-the-loop design; champion network; validation publications
 - 3. Data Access Limitations** - *Risk:* Hospital data sharing restrictions - *Mitigation:* Federated learning; on-premise deployment options; IRB-approved research protocols
 - 4. Pharma Budget Cycles** - *Risk:* Long enterprise sales cycles - *Mitigation:* Land with trial recruitment (urgent need); expand to broader services; success-based pricing options
 - 5. Patient Privacy Concerns** - *Risk:* Genetic data sensitivity - *Mitigation:* Patient-controlled sharing; differential privacy; HIPAA/GDPR compliance; transparent data use policies
-

The Vision: A World Where No Diagnosis Takes Years

Our 5-Year Moonshot Goals

- 2026:** Become the standard of care for undiagnosed patients at top 100 medical centers
- 2027:** Enable diagnosis of 50,000+ rare disease patients annually
- 2028:** Connect every rare disease patient to relevant clinical trials within 30 days
- 2029:** Achieve 90% diagnostic accuracy across all 7,000+ rare diseases
- 2030:** End the diagnostic odyssey — no patient waits more than 90 days for answers

The Future We're Building

Imagine a world where: - A parent's first concern about their child triggers intelligent screening - AI detects the subtle pattern that specialists might miss - Diagnosis happens in weeks, not years - Every treatment option and clinical trial is visible - Patients find others who share their journey - Researchers can find patients for life-saving trials - Cures are found because patients can be identified

This is the future RarePath AI is building. One patient at a time. One diagnosis at a time. One cure at a time.

Investment Highlights

Why RarePath AI Will Win

1. **Massive Underserved Market:** 400M patients, \$180B+ market, 95% without treatments
2. **Urgent Unmet Need:** Families desperate for answers, pharma desperate for patients
3. **Defensible Technology:** Proprietary models trained on 50M+ encounters
4. **Strong Network Effects:** Every patient improves the system
5. **Multiple Revenue Streams:** Pharma, health systems, patients, research
6. **Mission-Aligned Team:** Clinicians, researchers, and technologists united by purpose
7. **Regulatory Validation:** FDA breakthrough designation, CE marking
8. **Demonstrated Traction:** \$6.2M ARR, 340 medical centers, 89,000 patients

The Ask

Series A: \$18 Million

Use of Funds: - AI Platform Enhancement: 35% - Clinical Validation & Regulatory: 20% - Pharma Partnership Development: 20% - International Expansion: 15% - Team & Operations: 10%

Target Investors: - Healthcare-focused growth equity - AI/ML-focused technology investors - Strategic pharma corporate ventures - Impact-focused funds

Contact

RarePath AI

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Series A
\$18M at \$90M pre-money valuation

“Because every rare disease patient deserves answers.”

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